

Miguel Ángel Torres Montoya

Crime Economics · Urban Economics · Applied Econometrics · Spatial Methods
miguet2209@gmail.com | miguel.torres2@u.icesi.edu.co | +57 (316) 244-0219
GitHub | LinkedIn | Web-page

EDUCATION

Icesi University,	Colombia. Bachelor of Arts in Economics. GPA: 4.5/5.0. Expected 2027 Graduate-level Coursework: Quantitative and Computational Methods (4.8), Advanced Macroeconomics (in progress).
Berchmans School,	Cali, Colombia. High School Diploma (<i>Bachiller</i>). 2022

RESEARCH IN PROGRESS

Spatial Disaggregation of GDP and Informality in Bogotá.

With Eduard F. Martínez-González.

Spatially disaggregates Bogotá's GDP using nighttime lights, DANE 2018 census data, and mobility data; maps informality via XGBoost/Neural Networks.

Gender Disparities in the Colombian Criminal Justice System.

Undergraduate Thesis. Supervised by Eduard F. Martínez-González and Daniel Mejía-Londoño.

Examines gender differences in criminal sentencing in Colombia's accusatory system exploiting a natural experiment.

Strategic Default in Credit Unions.

With Eduard F. Martínez-González.

Econometric analysis of strategic default behavior in Colombian credit unions, examining borrower incentives when multiple institutions are available.

Sentencing Process Analysis in the Colombian Accusatory System.

With Eduard F. Martínez-González and Daniel Mejía-Londoño.

Examines the full criminal justice pipeline from criminal notice to final court ruling, identifying procedural bottlenecks and sentencing disparities.

Construction of Economic Centres in Cali.

With Cámara de Comercio de Cali and Invest Pacific.

Delineates Cali's economic subcenters by sector using georeferenced firm-level data and a spatial activity index.

RESEARCH EXPERIENCE

2025 – present	Research Assistant, Center for Research in Economics and Finance (CIENFI), Universidad Icesi, Cali, Colombia. Supervisor: Eduard F. Martínez-González. Urbanization Prediction Model for Colombia.
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Landsat 8 imagery (NDBI, NDVI, NDWI) integrated with DANE 2018 census blocks; k-NN and XGBoost classifiers to predict urban extent across Colombia (2013–2025).

Spatial Disaggregation of GDP for Bogotá. (With the Secretaría de Desarrollo Económico de Bogotá)

Nighttime lights (Suomi-NPP, NOAA-20), TransMilenio mobility data, and DANE 2018 census. Findings featured in *El Tiempo*, February 2025.

Property Registry Revaluation Analysis for Cali. (With the Alcaldía de Cali)

Descriptive analysis of the *reavalúo catastral*, focusing on property owner claims following the reassessment.

Data cienfi.

Open data repository for Business Analytics courses at Universidad Icesi.

TEACHING EXPERIENCE

2025 – present	Teaching Monitor, Introduction to Business Analytics, Universidad Icesi, Cali, Colombia.
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AWARDS AND HONORS

2025	3 rd Place, Undergraduate Presentation Competition on the Economics of Crime, Universidad Icesi.
2024	2 nd Place, “Who Wants to Be a Millionaire?” Economics Edition, Universidad Icesi.
2023	1 st Place, Undergraduate Economic Debate Competition, Universidad Icesi.

TECHNICAL SKILLS

Programming:	R (Advanced), Python (Intermediate).
Geospatial:	Google Earth Engine (rgee), QGIS.
Other Tools:	Excel (Advanced), LaTeX, Git, PowerPoint, Canva.
Methods:	Machine Learning (XGBoost, k-NN, Neural Networks), Spatial Econometrics, Remote Sensing, Statistical Modeling, Data Visualization.

LANGUAGES

Spanish:	Native.
English:	Professional Working Proficiency (C1).